ELECTRICAL—BATTERY DRAIN DIAGNOSIS—SERVICE TIPS	Article No. 02-9-5

FORD: 2002 THUNDERBIRD

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ISSUE

Some vehicles may exhibit excessive electrical system current draw which could result in a discharged battery and a no crank condition.

ACTION

Use the following Service Tips to monitor the vehicle electrical system for excessive battery current drain and repair as needed. Refer to the following Service Information Tips for details.

SERVICE INFORMATION

Having a firm grasp on the design and normal operation of the electrical system will aid in properly diagnosing and repairing any faults with electrical components and/or circuitry.

When the vehicle is at rest (ignition key turned off and removed), some amounts of battery current drain will be present since many of the modules and electronic components in the vehicle require this to sustain memory and system monitoring. The Powertrain Control Module, Antilock Brake Module, Electronic Automatic Temperature Control Module, Front Electronics Module, Rear Electronics Module, Driver Door Module, Instrument Cluster Module, Roof Opening Panel Module, Heated Seat Module, Driver Seat Module, Cell Phone Module and Vehicle Emergency Message System Module all have power supplied at all times for this purpose. Many of these modules monitor activity and communicate with each other through the Standard Corporate Protocol (SCP) Link.

When the ignition key is first shut off and removed, the vehicle is exited and the doors are closed, the electrical system goes into “stand-by” mode, where the Rear Electronics Module (REM) and the Front Electronics Module (FEM) keep the System Switched Power (SSP) relays activated. After a minimum of 45 minutes, the REM and FEM will power down to prevent battery discharge. Power down mode occurs when the REM and FEM “time out” and release the ground on all 4 (four) of the SSP relays. The SSP relays are wired in a fashion in which they are either all on or all off. In other words, the SSP relays cannot be operated independent of each other. If power down does not occur, because of a system fault or errant SCP Link activity, the REM and FEM will remain energized along with the SSP relays and the other electrical devices that they provide power. The REM and FEM provide power and ground to devices such as exterior/interior lighting and many of the body modules in the vehicle. The Wiring Diagrams Manual may be referenced for additional information on the SSP relays, the REM and FEM, in Sections 13 & 59.

Once the REM and FEM enter power down mode they will not energize the SSP relays until they are commanded to “wake up” by one or more of the following commands:

- Key Fob Activity (Any Command)
- Door Ajar (Any Door)
- Luggage Compartment Ajar Switch
- Drivers Power Seat Switch
- Power Tilt/Telescopic Steering Column
- Headlamps/Park Lamps On
- Key In Ignition
- Key On Ignition
- Anti-Theft Hood Switch
- Brake Pedal Position Switch
- Memory Seat Switch

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BATTERY DRAIN DIAGNOSIS

In order to properly diagnose a battery drain concern, the following procedures must be followed exactly as stated to produce the most accurate results.

1. Ask the customer how long the vehicle must sit before the battery goes dead. The vehicle is equipped with a 750 Amp Hour battery from the factory. If the battery is in proper working order, it will be able to withstand a 1 Amp draw for 48 hours without dropping the battery voltage below 10.5V. This information may help determine the amount of current drain present. For example, if a vehicle with a fully charged battery will not start after an over night soak, it can be assumed the battery current drain is greater than 4 amps. This knowledge may aid in diagnosis.
2. Check for any Aftermarket equipment installed in the vehicle. Ask the customer if they removed any Aftermarket equipment before bringing vehicle in for service. All Aftermarket equipment remaining should be removed before proceeding with this test.
3. Ask the customer if any electrical component(s) is inoperative or malfunctioning. In many cases a draining battery concern can usually be traced back to a component that has experienced operational trouble.
4. Using an appropriate scan tool, check all available modules in the vehicle for both On-Demand and Continuous Memory Diagnostic Trouble Codes (DTCs). Repair any On-Demand DTC's before continuing with this article. Record any Continuous Memory DTCs which may aid in future diagnosis.
5. Check that the vehicle battery is properly charged and that the charging system is operating correctly. Refer to Workshop Manual Section 414-00.
6. Install a quick disconnect style Battery Saver/Vehicle Anti-Theft Switch between the negative battery post and cable, such as Littlefuse part number ATD 100 BP or Moroso part number 74103.
7. Drive the vehicle for at least 20 minutes over 48 km/h (30 mph) to reach full engine operating temperature and to turn on and activate vehicle systems. Make sure all electrical components operate correctly. i.e. horn, audio system, exterior/interior lighting, climate control, etc.
8. Open the Left Front and Right Front door and engage the latches so the vehicle interior can be accessed with out disturbing the modules during this test.
9. Lift the hood and compress the Anti-Theft Hood Switch with a clamp (if equipped).
10. Open the trunk lid and unplug the electrical connection to the Luggage Compartment Lamp. Lift and secure the trunk floor panel hook to the trunk opening pinch weld. The Battery and the Rear Power Distribution Box are now easily accessible.

NOTE
DO NOT ALLOW BATTERY POWER TO BE DISCONNECTED FROM THE VEHICLE AT ANY TIME DURING THIS TEST. IF BATTERY DISCONNECT OCCURS, THE ENTIRE TEST MUST BE PERFORMED AGAIN STARTING AT STEP 5.

11. Use a Multimeter (DVOM) with a MIN/MAX feature to measure the battery current drain. The DVOM must be capable of reading DC milliamps (minimum 1mA) and have 10 amp - fused capacity.
12. Use the following procedure to measure current drain.
    a. Turn on the DVOM and select DC AMPS.
    b. Attach a large alligator clip on each side of the Battery Saver/Anti-Theft Switch Terminals.
    c. Connect the DVOM in series by securing the leads to the alligator clips on the Battery Saver/Anti-Theft terminals.
    d. Loosen the knob on the Battery Saver/Anti-Theft Switch to break the contact between the negative battery terminal and cable.

NOTE
DO NOT USE A TEST LIGHT OR A VOLT METER AS A CHECK FOR EVIDENCE OF CURRENT

FLOW. A TEST LIGHT WILL NOT ACCURATELY DETERMINE THE AMOUNT OF CURRENT FLOW.

13. Wake the electrical system up by depressing the lock or unlock button on the key fob. Observe the reading on the DVOM. The current flow will "jump" momentarily then after 1-2 minutes settle to the level listed below (a). Actual readings may vary slightly depending on vehicle equipment, battery condition, ambient temperature and DVOM:

    a. .85 amps (850mA) or less (Stand By Mode)

    b. .035 amps (35mA) or less after 45 minutes minimum (Power Down Mode)

14. During Stand By Mode, current drain readings of 850mA or less indicate the vehicle is performing normally. However, the concern may be intermittent in nature. Set the MIN/MAX feature on the DVOM and periodically check the meter for evidence of excessive current drain. The MIN/MAX feature will record the highest measured current reading.

15. An electrical system that is operating normally will power down to 35mA or less after 45 minutes minimum. However, it may not be necessary to wait for Power Down Mode to begin diagnosing the source of excessive battery current drain. For example: A current drain reading of 2.5 amps is observed during Stand By Mode. Knowing a normal system will measure 850mA current drain, we can estimate the suspect component is consuming 1.65 amps from the system.

16. With the Wiring Diagrams Manual and Owner's Guide easily accessible, begin to track the 1.65 amps excess drain by removing fuses one-at-a-time until the battery current drain drops to an acceptable level. Once the problem fuse is found, use the EVTM to guide you through the schematic identifying all potential components or circuits. The complexity of the vehicle makes it impossible to diagnose the source of current drain without the appropriate model year Wiring Diagrams Manual.

17. Start diagnosis in the Rear Power Distribution Box (RPDB) located in the trunk. Remove the smaller amperage mini-fuses, listed below by vehicle, one-at-a-time while checking the meter for current drain drop. If current drain does not drop down to 850mA, leave the fuse out and move on to the next. Continue to remove mini-fuses until all are removed from the RPDB. If the current drain does drop to an acceptable level, proceed to step 20. If the current drain in unaffected, proceed to step 18.

    a. Lincoln LS - Fuses 4.01 through 4.18

    b. Thunderbird - Fuses 4.17 through 4.41

NOTE

REMOVING AND REINSTALLING FUSES 2.20 (LS & THUNDERBIRD) & 4.14 (LS ONLY), DURING THE FOLLOWING TESTS WILL PROMPT THE SYSTEM TO "WAKE UP". THE CURRENT DRAIN ON THE METER WILL JUMP MOMENTARILY FOR 1-2 MINUTES THEN SLOWLY DECREASE TO PREVIOUSLY MEASURED LEVELS.

NOTE

DO NOT REMOVE THE LARGE AMPERAGE CARTRIDGE FUSES OR ANY RELAYS FROM THE RPDB. BY DOING SO POWER WILL BE LOST TO MULTIPLE FUSES, RELAYS AND COMPONENTS ALL AT THE SAME TIME. THIS TEST IS ATTEMPTING TO LOCATE THE BATTERY CURRENT DRAIN AS CLOSE TO THE COMPONENT LEVEL AS POSSIBLE.

NOTE

DO NOT REMOVE FUSES 4.23, 4.24, 4.27, 4.32 (LS) - 4.10, 4.15, 4.27, 4.28 (THUNDERBIRD) OR SSP RELAYS 1, 2, 3, 4 DURING STAND BY MODE. REMOVING THESE FUSES AND RELAYS WILL LESSEN BATTERY CURRENT DRAIN BECAUSE YOU HAVE REMOVED POWER TO THE REM & FEM DURING STAND BY MODE. DO NOT FALSELY DIAGNOSE A REM OR FEM PROBLEM AS A RESULT OF THIS ACTIVITY.

18. If the current drain is not discovered in the RPDB move to the Central Junction Box (CJB), located behind the Right Interior Kick Panel. Remove the mini-fuses one-at-a-time, 2.01 through 2.35, while checking the meter for current drain drop. If current drain does not drop down to 850mA, leave the fuse out and move on to the next. If the current drain does drop, proceed to Step 20.

19. If the current drain cannot be located in the RPDB or CJB, continue to the Auxiliary Junction Box (AJB) located in the engine compartment. All fuses in the AJB are Hot At All Times. Start by removing the mini-fuses 1.01 through 1.12. If current drain does not drop down to 850mA, leave the fuses out and move on to cartridge fuses 1.13 through 1.24. If current drain does not drop to 850mA, remove the relays from the AJB and check for current drain drop. If the current drain does drop, proceed to Step 20.

20. If the excess current drain is discovered when removing any of these fuses or relays, use the Wiring Diagrams Manual, Section 13, to determine the component and/or mini-fuses that are wired downstream in the circuit. Remove those mini-fuses or disable those components to help determine the root cause of the current drain. Use these findings to repair the vehicle as needed and repeat the process starting at Step 7 to verify the repair.

NOTE
IF IT'S DETERMINED THAT A MODULE IS THE SOURCE OF THE CURRENT DRAIN, SELF TEST THE MODULE WITH A SCAN TOOL FOR CODES. ALSO CHECK THE MODULES AVAILABLE PIDS INDEX FOR POSSIBLE ERRANT INPUTS THAT MAY BE CAUSING THE MODULE TO OPERATE ABNORMALLY.

By following this procedure, we have eliminated all potential sources of current drain while not disturbing SSP relays 1 through 4 or power to the REM & FEM. Continue to Step 21.

21. Cycle the key fob unlock button several times to make certain the vehicle is "awake". Leave the vehicle unattended and observe the battery drain after 45 minutes minimum. The vehicle should have entered Power Down Mode by this time which drops the current drain to 35mA or less.

22. If the current drain is greater than 35mA, reach inside vehicle and turn on the Left Front Map Lamp (LS - button on the dome lamp assembly, Thunderbird - button on the rear view mirror). Does the lamp illuminate?
    - Yes - The FEM & REM are awake and the SSP relays are active. Continue to next section of this TSB.
    - No - The FEM & REM have Powered Down. Return to Step 4 and repeat diagnosis.

23. If the current drain is 35mA or less the vehicle is operating normally at this time. However, the concern may be intermittent in nature. Continue with the next section of this article.

Errant activity or a false "wake up" message present in the SCP Link will not allow the system to power down, or may wake up a system that has been Powered Down. This condition may be intermittent in nature.

USING THE NGS TO CHECK FOR UNWANTED ACTIVITY

This section advises how to use the NGS tester to check the SCP Link for this type of unwanted activity. The NGS Service Card is equipped with a SCP Link Monitor Tool which allows the Technician the ability to monitor the communication, or traffic, between all modules equipped in the vehicle that use the SCP Link. Many of the vehicles functions are accomplished as a result of the communication between modules.

NOTE
THE NGS WILL CONSUME 507mA AT 12.6V. IF MEASURING CURRENT DRAIN WITH THE NGS INSTALLED, 507mA MUST BE SUBTRACTED FROM THE CURRENT DRAIN DISPLAYED ON THE DVOM TO DETERMINE THE ACTUAL AMOUNT.

1. Using NGS Service Function Card version 10.1 or higher, select SCP LINK MONITOR TOOL from the main menu.

2. With the ignition key in the OFF position, connect the NGS to the 16-pin Data Link Connector.

3. Select the SCP LINK MONITOR TOOL from the main menu and press the START key.

4. At this point, a normally operating vehicle should have no messages present in the SCP Link. If no messages are present, leave the NGS connected to the vehicle and periodically check for SCP activity. If the concern is intermittent, it may take some time before the suspect component behaves in an undesirable fashion. If a message is eventually observed, continue with Step 5 in this section. Because of the excessive drain caused by the NGS, it may be necessary to connect a battery charger to prevent a dead battery.

5. If messages are present, press the STOP key. Scroll through the messages on the screen using the scroll dial. The following information will be displayed in order on the NGS. Match this information with the data on the Message List found at the end of this article.
   - **FNCT** (FUNCTION): Displays the two digit ID that represents a category of function to be carried out by the module(s) receiving the message. Match this ID with the Function column on the message list.
   - **SRC** (SOURCE): Displays the module acronym responsible for sending the SCP message. Match the module Source displayed on the NGS to that on the Source column of the message list.
   - **DATA 01 - 06:** Displays more specific information related to the function. Match these characters with the data on columns 1-7 of the message list, along with the source observed, to define the message being viewed on the NGS.
   - **FLTR** (FILTER): Displays the current status of the FILTER function. You may manipulate the NGS to capture messages that are sent from a particular or chosen module (Source) or Function.

Compare the messages received on the NGS to the Message List tables at the end of this article. These will help determine the Function, Source and Name of the message read from the SCP Link. The Message List tables provide all possible SCP Link messages listed in order by Source. Most messages are the result of an input to one module being transmitted to another module that carries out a vehicle function. Messages may contain information that commands a module to perform a function, while other messages may provide information related to vehicle status that will allow or not allow a function to be performed. Depending on the command, a module may be able to accommodate the command entirely on its own or it may require another module to operate in a certain fashion to achieve the desired initial command. This is all communicated through the SCP Link.

For example, by turning on the headlamp switch the ICM receives the command (input) from the switch to activate all exterior lamps. The ICM does not directly control the exterior lamps, but instead will send a message out over the SCP link telling the FEM and REM that the customer has requested all exterior lamps be turned on. The FEM will receive this message and provide a ground activating all the lamps it has within its control in the front of the vehicle. Simultaneously, the REM will receive the same message over the SCP link requesting that all lamps in the rear of the vehicle associated with headlamp operation be activated.

NOTE

DO NOT PREMATURELY DETERMINE THAT THE SOURCE MODULE IDENTIFIED IS THE CAUSE OF THE BATTERY DRAIN CONDITION. IN MOST CASES, THE MESSAGE CAN BE TRACED TO A FAULTY COMPONENT SENDING A FALSE INPUT ON THE SCP LINK. THIS CONDITION MAY NOT ALLOW A MODULE TO POWER DOWN. IF THE ERRANT MESSAGE OCCURS AFTER POWER DOWN, IT MAY WAKE-UP A SYSTEM THAT HAS POWERED DOWN.

6. The following function keys are active when monitoring the SCP Link:
   - **RESET:** Clears the screen while the monitor is running
   - **FLTR:** Turns the filter feature ON or OFF as indicated by the FLRT display in the upper right screen
   - **SETUP:** When SETUP is pressed, the filter setup screen will appear. This will allow the operator to specify the capture of only certain messages by either its Source (particular module) or Function.
   - **START:** The START key will start the NGS in monitor mode, and display all traffic on the link. The START key will now be displayed as STOP. Pressing the STOP key will stop the monitor from reading any further traffic on the SCP link.

NOTE

PRESSING THE START KEY AGAIN WILL CLEAR THE SCREEN OF ALL CAPTURED MESSAGES.

NOTE

THE NGS WILL CONTINUOUSLY CAPTURE MESSAGES BUT WILL ONLY DISPLAY UP TO 100 MESSAGES AT ONE TIME. ONCE 100 MESSAGES ARE STORED, THE FIRST

MESSAGE WILL BE DELETED FROM THE LIST. THE MONITOR OPERATES ON THE "FIRST IN, FIRST OUT" PRINCIPAL.

NOTE
THE POWERTRAIN CONTROL MODULE (PCM) AND ANTI-LOCK BRAKE MODULE (ABS) SHOULD NOT DISPLAY ANY FORM OF MESSAGE ACTIVITY WITH THE KEY IN THE OFF POSITION. IF ANY PCM OR ABS MESSAGES ARE DETECTED AT THIS TIME, CHECK FOR VOLTAGE ON CIRCUITS WHICH SHOULD ONLY HAVE POWER WHEN THE KEY IS ON OR IN THE CRANK POSITION.

USING THE NGS TO CAPTURE MESSAGES FROM A PARTICULAR MODULE

If needed, the NGS may be manipulated to capture messages that are sent from a particular module (Source) or Function. To setup the FLTR feature to capture only messages sent by a module use the following procedure:

1. Enter the SETUP menu and press the SRC key.
2. Use the scroll to select the desired module and press Trigger.
3. Press the Cancel button to return to the monitor screen.
4. Press the FLTR button to toggle the FLTR display in the upper right screen to on.
5. The NGS will now display only messages sent by the module that you specified in the SETUP menu.

USING THE NGS TO CAPTURE MESSAGES RELATED TO CERTAIN FUNCTIONS

To setup the FLTR feature to capture only messages related to certain Functions use the following procedure:

1. Enter the SETUP menu and press the FNCT key.
2. Use the scroll to select the desired function from the message list and press Trigger.
3. Press the Cancel button to return to the monitor screen.
4. Press the FLTR button to toggle the FLTR display in the upper right screen to on.
5. The NGS will now display only messages relating the Function you specified in the SETUP menu.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 203000, 203100, 601300, 603300

Module Acronym	Module Name
ABS	Anti Lock Brake
ACM	Audio Control Module
ADCM	Adaptive Dampening Control Module
DDM	Driver Door Module
DEATC	Dual Electronic Automatic Temperature Control
DSM	Driver Seat Module
GEM	Generic Electronic Module
IC	Instrument Cluster
IVD	Interactive Vehicle Dynamics (ABS)
MC	Message Center
NAV	Navigation System
PCM	Powertrain Control Module
REM	Rear Electronics Module
TC	Traction Control (ABS)
VACM	Voice Audio Control Module
VEMS	Vehicle Emergency Messaging System

TB-6591-A

Figure 1 - Article 02-9-5

Function	Source	DATA							MESSAGE NAME
		1	2	3	4	5	6	7	
29	ABS	03	XX	XX	XX	XX	XX	XX	Vehicle Speed - Driven and Undriven Wheels - High Res
33	ABS	3B	XX						Brake System Configuration Status
88	ABS	19							Brake System Failed Telltale Command - Off
88	ABS	88							Anti-Lock Brake System Failed Telltale Command - On
88	ABS	99							Brake System Failed Telltale Command - On
88	ABS	08							Anti-Lock Brake System Failed Telltale Command - Off
3B	ABS	FB							Request Transmission Configuration Status
7B	ABS	02	XX						Odometer Rolling Count Status
75	ACM	41							Request Delayed Accessory Command
87	ACM	C4							Request Ignition Switch Position w/ Initialize Status
AD	ACM	01	XX	XX					Remote Control #1 Button Status
DF	ACM	52							Request Backlighting Intensity & Dimming Curve w/ Headlamps Command
EA	ACM	10	XX	XX	XX	XX	XX	XX	Download Block to Display Command
EA	ACM	11							Terminate Display Definition Command
EA	ACM	30	XX						Display Access Display String Command - Clear Display
EA	ACM	31							Display Access Terminate Command
33	ADCM	E2							Request Brake Lamp Pedal Switch Status
59	ADCM	06	XX	XX					Suspension System State Status
33	DDM	E0							Request Parking Brake Switch Status
74	DDM	81							Delayed Accessory Command - On
74	DDM	01							Delayed Accessory Command - Off
87	DDM	C4							Request Ignition Switch Position w/ Initialize Status
87	DDM	C5							Request Key-in-Ignition Status
92	DDM	13							Remote Panic Command - Disable(d)
92	DDM	93							Remote Panic Command - Enable(d)
96	DDM	83							Horn Configuration 3 Command - Enable(d)
B4	DDM	89							Memory Features 1 Command - Set / Save
B4	DDM	09							Memory Features 1 Command - Recall
B4	DDM	0A							Memory Features 2 Command - Recall
B4	DDM	8A							Memory Features 2 Command - Set / Save
B4	DDM	A0							Memory Features Recall Cancel Command - Yes (True)
B5	DDM	10	XX	XX	XX	XX			Memory Feature Menu Status
C2	DDM	81	1E						Passenger's Mirror Up Motion Command - Enable(d)
C2	DDM	82	1E						Passenger's Mirror Down Motion Command - Enable(d)
C2	DDM	83	1E						Passenger's Mirror Right Motion Command - Enable(d)
C2	DDM	84	1E						Passenger's Mirror Left Motion Command - Enable(d)
C2	DDM	04	1E						Passenger's Mirror Left Motion Command - Disable(d)
C4	DDM	10	00						All Door Lock Motor Enable Command - Disable(d)
C4	DDM	81	00						All Door Lock Command - Lock(ed)
C4	DDM	87	00						All Super / Double Door Lock Command - Lock(ed)
C4	DDM	89	00						All Remote Door Lock Command - Lock
C4	DDM	01	00						All Remote Door Lock Commmand - Unlock(ed)
C4	DDM	09	00						All Remote Door Lock Command - Unlock
C5	DDM	06	22	XX					Driver's Front Door Lock Cylinder State Status
C5	DDM	A0	22						Driver's Front Door Lock Switch Status - Active
C5	DDM	A1	22						Driver's Front Door Unlock Switch Status - Active
C6	DDM	81	34						Decklid Door Open Command - Yes (True)
C7	DDM	E1	34						Request Decklid Door Ajar Switch Status
C7	DDM	E1	22						Request Driver's Front Door Ajar Swtich Status
C7	DDM	E1	2A						Request Driver's Rear Door Ajar Switch Status
C7	DDM	E1	14						Request Hood Door Ajar Swtich Status
C7	DDM	E1	26						Request Passenger's Front Door Ajar Switch Status
C7	DDM	E1	2E						Request Passenger's Rear Door Ajar Switch Status
CA	DDM	81	2A						Driver's Rear Window Open Command - Enable(d)
CA	DDM	81	26						Passenger's Front Window Open Command - Enable(d)
CA	DDM	81	2E						Passenger's Rear Window Open Command - Enable(d)
CA	DDM	82	2A						Driver's Rear Window Close Command - Enable(d)
CA	DDM	82	26						Passenger's Front Window Close Command - Enable(d)
CA	DDM	82	2E						Passenger's Rear Window Close Command - Enable(d)
CA	DDM	84	2A						Driver's Rear Master Controller Window Open Command - Enable(d)
CA	DDM	84	24						Front Sun Roof Master Controller Window Open Command - Enable(d)
CA	DDM	84	26						Passenger's Front Master Controller Window Command - Enable(d)
CA	DDM	84	2E						Passenger's Rear Master Controller Window Command - Enable(d)
CA	DDM	85	2A						Driver's Rear Master Controller Window Close Command - Enable(d)

TB-6592-A

Figure 2 - Article 02-9-5

Function	Source	DATA							MESSAGE NAME
		1	2	3	4	5	6	7	
CA	DDM	85	24						Front Sun Roof Master Controller Window Close Command - Enable(d)
CA	DDM	85	26						Passenger's Front Master Controller Window Close Command - Enable(d)
CA	DDM	85	2E						Passenger's Rear Master Controller Window Close Command - Enable(d)
CA	DDM	01	2A						Driver's Rear Window Open Command - Disable(d)
CA	DDM	01	26						Passenger's Front Window Open Command - Disable(d)
CA	DDM	01	2E						Passenger's Rear Window Open Command - Disable(d)
CA	DDM	01	2A						Driver's Rear Window Close Command - Disable(d)
CA	DDM	02	26						Passenger's Front Window Close Command - Disable(d)
CA	DDM	02	2E						Passenger's Rear Window Close Command - Disable(d)
CA	DDM	04	2A						Driver's Rear Master Controller Window Open Command - Disable(d)
CA	DDM	04	24						Front Sun Roof Master Controller Window Open Command - Disable(d)
CA	DDM	04	26						Passenger's Front Master Controller Window Open Command - Disable(d)
CA	DDM	04	2E						Passenger's Rear Master Controller Window Open Command - Disable(d)
CA	DDM	05	2A						Driver's Rear Master Controller Window Close Command - Disable(d)
CA	DDM	05	24						Front Sun Roof Master Controller Window Close Command - Disable(d)
CA	DDM	05	26						Passenger's Front Master Controller Window Close Command - Disable(d)
CA	DDM	05	2E						Passenger's Rear Master Controller Window Close Command - Disable(d)
CB	DDM	23	28						All Rear Window Lockout Swtich Status - Inactive
CB	DDM	A3	28						All Rear Window Lockout Switch Status - Active
FE	DDM	82							Network Bus Wake-up Command - Yes (True)
14	DEATC	84							A/C Clutch Required Command - Yes (True)
14	DEATC	04							A/C Clutch Required Command - No (False)
49	DEATC	D0							Request Engine Coolant Fluid Temperature Status
DF	DEATC	52							Request Backlighting Intensity & Dimming Curve w/ Headlamps Command
E7	DEATC	21	34						Rear Windshield Electric Defrost Switch Status - Inactive
E7	DEATC	81	1C						Front Windshield Electric Defrost Status - On
E7	DEATC	01	1C						Front Windshield Electric Defrost Status - Off
E7	DEATC	A1	34						Rear Windshield Electric Defrost Switch Status - Active
F3	DEATC	10	XX						Outside Air Temperature Status
F9	DEATC	D6							Request Engine Off Elapsed Time Status
87	DSM	C4							Request Ignition Switch Position w/ Initialize Status
87	DSM	C5							Request Key-in-Ignition Status
B4	DSM	A0							Memory Features Recall Cancel Command - Yes (True)
B5	DSM	D0							Request Memory Feature Menu Status
FE	DSM	82							Network Bus Wake-up Command - Yes (True)
33	GEM	20							Parking Brake Switch Status - Inactive
33	GEM	A0							Parking Brake Switch Status - Active
33	GEM	E2							Request Brake Lamp Pedal Switch Status
74	GEM	84							Battery Saver Command - On
74	GEM	04							Battery Saver Command - Off
75	GEM	41							Request Delayed Accessory Command
87	GEM	C4							Request Ignition Switch Position w/ Initialize Status
87	GEM	C5							Request Key-in-Ignition Status
88	GEM	10							Low Washer Fluid Telltale Command - Off
88	GEM	20							Low Brake Fluid Telltale Command - Off
88	GEM	90							Low Washer Fluid Telltale Command - On
88	GEM	A0							Low Brake Fluid Telltale Command - On
92	GEM	02	XX						Vehicle Security Visual Indicator Mode Command
96	GEM	91							Chime Configuration 1 Command - Enable(d)
3A	GEM	25	XX						Transmission Performance Mode Command
3B	GEM	E5							Request Transmission Performance Mode Status
4B	GEM	33							Engine Oil Fluid Pressure Low Status - No (False)
4B	GEM	B3							Engine Oil Fluid Pressure Low Status - Yes (True)
B5	GEM	D0							Request Memory Feature Menu Status
C7	GEM	21	22						Driver's Front Door Ajar Switch Status - Inactive
C7	GEM	21	14						Hood Door Ajar Switch Status - Inactive
C7	GEM	21	26						Passenger's Front Door Ajar Switch Status - Inactive
C7	GEM	A1	22						Driver's Front Door Ajar Switch Status - Active
C7	GEM	A1	14						Hood Door Ajar Switch Status - Active
C7	GEM	A1	26						Passenger's Front Door Ajar Switch Status - Active
C7	GEM	E1	34						Request Decklid Door Ajar Switch Status
C7	GEM	E1	2A						Request Driver's Rear Door Ajar Switch Status
C7	GEM	E1	2E						Request Passenger's Rear Door Ajar Switch Status
CB	GEM	A1	26						Passenger's Front Window Open Switch Status - Active

TB-6593-A

Figure 3 - Article 02-9-5

Function	Source	DATA 1	2	3	4	5	6	7	MESSAGE NAME
D9	GEM	85	09						Left Front Turn Lamp OK Status - Yes (True)
D9	GEM	85	29						Left Side Mid Vehicle Turn Lamp OK Status - Yes (True)
D9	GEM	85	0F						Right Front Turn Lamp OK Status - Yes (True)
D9	GEM	85	2F						RIght Side Mid Vehicle Turn Lamp OK Status - Yes (True)
D9	GEM	05	09						Left Front Turn Lamp OK Status - No (False)
D9	GEM	05	29						Left Side Mid Vehicle Turn Lamp OK Status - No (False)
D9	GEM	05	0F						Right Front Turn Lamp OK Status - No (False)
D9	GEM	05	2F						Right Side Mid Vehicle Turn Lamp OK Status - No (False)
DA	GEM	84	38						All Rear Park Lamp Command - On
DA	GEM	85	00						All Turn Lamp Command - On
DA	GEM	04	38						All Rear Park Lamp Command - Off
DA	GEM	05	00						All Turn Lamp Command - Off
DB	GEM	41	00						Request All Headlamp Command
DB	GEM	44	00						Request All Park Lamp Command
DB	GEM	46	08						Request All Front High Beam Lamp Command
DB	GEM	49	08						Request All Front Fog Lamp Command
DB	GEM	86	08						All Front High Beam Lamp Status - On
DB	GEM	89	08						All Front Fog Lamp Status - On
DB	GEM	06	08						All Front High Beam Lamp Status - Off
DB	GEM	09	08						All Front Fog Lamp Status - Off
DF	GEM	52							Request Backlighting Intensity & Dimming Curve w/ Headlamps Command
DF	GEM	E1	00						Request All Courtesy Lamp Switch Status
FE	GEM	82							Network Bus Wake-up Command - Yes (True)
29	IC	10							Vehicle Overspeed Warning Status - No
29	IC	90							Vehicle Overspeed Warning Status - Yes
33	IC	E0							Request Parking Brake Switch Status
49	IC	30							Engine Coolant Fluid Temperature High Status - No (False)
49	IC	B0							Engine Coolant Fluid Temperature High Status - Yes (True)
49	IC	D0							Request Engine Coolant Fluid Temperature Status
49	IC	E6							Request Fail Safe Cooling Mode Status
63	IC	C2							Request Vehicle Speed Control Set Speed Status
73	IC	D4							Request Charging System State Status
75	IC	44							Request Battery Saver Command
83	IC	12	XX						Fuel Level - Percent Status
87	IC	84	XX						Ignition Switch Position w/ Initialize Status - Yes (True)
87	IC	85							Key-in-Ignition Status -Yes (True)
87	IC	04	XX						Ignition Switch Position w/ Initialize Status - No (False)
87	IC	05							Key-in-Ignition Status - No (False)
89	IC	48							Request Anti-Lock Brake System Failed Telltale Command
89	IC	50							Request Low Washer Fluid Telltale Command
89	IC	51							Request Traction Control Active Telltale Command
89	IC	56							Request IVD Active Telltale Command
89	IC	59							Request Brake System Failed Telltale Command
89	IC	60							Request Low Brake Fluid Telltale Command
89	IC	70							Request Check Engine (MIL) Telltale Indicator Mode Command
92	IC	1B							Steering Column Lock Command - Unlock
92	IC	1C							Steering Column Lock Enable Command - Off
92	IC	1D							Steering Column Lock Warning Command - No
92	IC	9B							Steering Column Lock Command - Lock
92	IC	9C							Steering Column Lock Enable Command - On
92	IC	9D							Steering Column Lock Warning Command - Yes
96	IC	81							Horn Configuration 1 Command - Enable(d)
96	IC	01							Horn Configuration 1 Command - Disable(d)
2B	IC	C6							Request Traction Control System State Status
3B	IC	21	XX						Transmission System State Status
3B	IC	C3							Request Transmission Actual Gear Position w/ Shift in Progress Status
3B	IC	E1							Request Transmission System State Status
3B	IC	E6							Request Transmission Operating Mode Status
3B	IC	FB							Request Transmission Configuration Status
4B	IC	F3							Request Engine Oil Fluid Pressure Low Status
B4	IC	A0							Memory Features Recall Cancel Command - Yes (True)
B5	IC	10	XX	XX	XX	XX			Memory Feature Menu Status
B5	IC	D0							Request Memory Feature Menu Status
C7	IC	E1	34						Request Decklid Door Ajar Switch Status
C7	IC	E1	22						Request Driver's Front Door Ajar Swtich Status

TB-6594-A

Figure 4 - Article 02-9-5

Function	Source	DATA 1	2	3	4	5	6	7	MESSAGE NAME
C7	IC	E1	2A						Request Driver's Rear Door Ajar Switch Status
C7	IC	E1	26						Request Passenger's Front Door Ajar Switch Status
C7	IC	E1	2E						Request Passenger's Rear Door Ajar Switch Status
D9	IC	C2	39						Request Left Rear Tail Lamp OK Status
D9	IC	C2	3F						Request Right Rear Tail Lamp Ok Status
D9	IC	C3	39						Request Left Rear Brake Lamp OK Status
D9	IC	C3	3F						Request Right Rear Brake Lamp OK Status
D9	IC	C5	09						Request Left Front Turn Lamp OK Status
D9	IC	C5	39						Request Left Rear Turn Lamp OK Status
D9	IC	C5	29						Request Left Side Mid Vehicle Turn Lamp OK Status
D9	IC	C5	0F						Request Right Front Turn Lamp OK Status
D9	IC	C5	3F						Request Right Rear Turn Lamp OK Status
D9	IC	C5	2F						Request Right Side Mid Vehicle Turn Lamp OK Status
DA	IC	10	XX						Autolamp Delay Command - Disable(d)
DA	IC	81	00						All Headlamp Command - On
DA	IC	84	00						All Park Lamp Command - On
DA	IC	85	00						All Turn Lamp Command - On
DA	IC	85	01						Left Side Turn Signal Turn Lamp Command - On
DA	IC	85	07						Right Side Turn Signal Turn Lamp Command - On
DA	IC	86	08						All Front High Beam Lamp Command - On
DA	IC	89	08						All Front Fog Lamp Command - On
DA	IC	89	38						All Rear Fog Lamp Command - On
DA	IC	90	XX						Autolamp Delay Command - Enable(d)
DA	IC	01	00						All Headlamp Command - Off
DA	IC	04	00						All Park Lamp Command - Off
DA	IC	05	00						All Turn Lamp Command - Off
DA	IC	05	01						Left Side Turn Signal Turn Lamp Command - Off
DA	IC	05	07						Right Side Turn Signal Turn Lamp Command - Off
DA	IC	06	08						All Front High Beam Lamp Command - Off
DA	IC	09	08						All Front Fog Lamp Command - Off
DA	IC	09	38						All Rear Fog Lamp Command - Off
DB	IC	C6	08						Request All Front High Beam Lamp Status
DB	IC	C9	08						Request All Front Fog Lamp Status
DB	IC	C9	38						Request All Rear Fog Lamp Status
DE	IC	12	XX	XX					Backlighting Intensity & Dimming Curve w/ Headlamps Command - Off
DE	IC	92	XX	XX					Backlighting Intensity & Dimming Curve w/ Headlamps Command - On
DF	IC	21	00						All Courtesy Lamp Switch Status - Inactive
DF	IC	A1	00						All Courtesy Lamp Switch Status - Active
EB	IC	C8							Request English / Metric Display Status
F9	IC	17	XX	XX					Vehicle Power Connection - Elapsed Time Status
FE	IC	82							Network Bus Wake-up Command - Yes (True)
8	IVD	0C	XX	XX					Desired Axle Torque Command
29	IVD	03	XX	XX	XX	XX	XX	XX	Vehicle Speed - Driven and Undriven Wheels - High Res
33	IVD	3B	XX						Brake System Configuration Status
33	IVD	E2							Request Brake Lamp Pedal Switch Status
53	IVD	FB							Request Engine Configuration Status
88	IVD	16							IVD Active Telltale Command - Off
88	IVD	19							Brake System Failed Telltale Command - Off
88	IVD	88							Anti-Lock Brake System Failed Telltale Command - On
88	IVD	96							IVD Active Telltale Command - On
88	IVD	99							Brake System Failed Telltale Command - On
88	IVD	08							Anti-Lock Brake System Failed Telltale Command - Off
2B	IVD	06	XX	XX					Traction Control System State Status
3B	IVD	C3							Request Transmission Actual Gear Position w/ Shift in Progress Status
3B	IVD	CA							Request Clutch Pedal Position Status
3B	IVD	FB							Request Transmission Configuration Status
7B	IVD	02	XX						Odometer Rolling Count Status
DA	IVD	83	38						All Rear Brake Lamp Command - On
DA	IVD	03	38						All Rear Brake Lamp Command - Off
13	MC	C5							Request Electronic Throttle Control Warning Status
33	MC	FB							Request Brake System Configuration Status
39	MC	08	XX						Language Code Status
49	MC	E6							Request Fail Safe Cooling Mode Status
49	MC	F0							Request Engine Coolant Fluid Temperature High Status

TB-6595-A

Figure 5 - Article 02-9-5

Function	Source	DATA 1	2	3	4	5	6	7	MESSAGE NAME
73	MC	D4							Request Charging System State Status
82	MC								Fuel Flow (FR)
87	MC	C4							Request Ignition Switch Position w/ Initialize Status
89	MC	50							Request Low Washer Fluid Telltale Command
89	MC	60							Request Low Brake Fluid Telltale Command
96	MC	91							Chime Configuration 1 Command - Enable(d)
96	MC	92							Chime Configuration 2 Command - Enable(d)
2B	MC	C6							Request Traction Control System State Status
3B	MC	E1							Request Transmission System State Status
B5	MC	10	XX	XX	XX	XX			Memory Feature Menu Status
B5	MC	D0							Request Memory Feature Menu Status
C7	MC	E1	34						Request Decklid Door Ajar Switch Status
C7	MC	E1	22						Request Driver's Front Door Ajar Switch Status
C7	MC	E1	2A						Request Driver's Rear Door Ajar Switch Status
C7	MC	E1	26						Request Passenger's Front Door Ajar Switch Status
C7	MC	E1	2E						Request Passenger's Rear Door Ajar Switch Status
D9	MC	C2	39						Request Left Rear Tail Lamp OK Status
D9	MC	C2	3F						Request Right Rear Tail Lamp OK Status
D9	MC	C3	39						Request Left Rear Brake Lamp OK Status
D9	MC	C3	3F						Request Right Rear Brake Lamp OK Status
D9	MC	C5	09						Request Left Front Turn Lamp OK Status
D9	MC	C5	39						Request Left Rear Turn Lamp OK Status
D9	MC	C5	29						Request Left Side Mid Vehcle Turn Lamp OK Status
D9	MC	C5	0F						Request Right Front Turn Lamp OK Status
D9	MC	C5	3F						Request Right Rear Turn Lamp OK Status
D9	MC	C5	2F						Request Right Side Mid Vehicle Turn Lamp OK Status
DF	MC	52							Request Backlighting Intensity & Dimming Curve w/ Headlamps Command
EB	MC	12	XX						Terminate Display Confirmation Status - Reject
EB	MC	32	XX						Display Access Confirmation Status - Reject
EB	MC	88							English / Metric Display Status - Metric
EB	MC	92	XX						Terminate Display Confirmation Status - Accept
EB	MC	08							English / Metric Display - English
EB	MC	B2	XX						Display Access Confirmation - Accept
F9	MC	D7							Request Vehicle Power Connection- Elapsed Time Status
33	NAV	E0							Request Parking Brake Switch Status
39	NAV	08	XX						Language Code Status
87	NAV	C4							Request Ignition Switch Position w/ Initialize Status
AA	NAV	10	XX						Driver's Assistance Phone Command
DB	NAV	44	00						Request All Park Lamp Command
EB	NAV	12	XX						Terminate Display Confirmation Status - Reject
EB	NAV	32	XX						Display Access Confirmation Status - Reject
EB	NAV	92	XX						Terminate Display Confirmation Status - Accept
EB	NAV	B2	XX						Display Access Confirmation Status - Accept
9	PCM	22	XX	XX	XX	XX	XX	XX	Actual Axle Torque w/ Min & Max Available Torque Status
13	PCM	05	XX	XX					Electronic Throttle Control Warning Status
13	PCM	E0							Request Throttle Control Unit Fault Status
33	PCM	E0							Request Parking Brake Switch Status
33	PCM	E2							Request Brake Lamp Pedal Switch Status
49	PCM	10	XX						Engine Coolant Fluid Temperature Status
49	PCM	26	XX						Fail Safe Cooling Mode Status
49	PCM	32							Engine Coolant Low Level Status - No (False)
49	PCM	B2							Engine Coolant Low Level Status - Yes (True)
53	PCM	3B	XX	XX	XX	XX			Engine Configuration Status
63	PCM	81							Vehicle Speed Control System Status - On
63	PCM	82	XX						Vehicle Speed Control Set Speed Status - Enable(d)
63	PCM	01							Vehicle Speed Control System Status - Off
63	PCM	02	XX						Vehicle Speed Control Set Speed Status - Disable(d)
73	PCM	14	XX						Charging System State Status
83	PCM	FE							Request Fuel I/O Status
87	PCM	C4							Request Ignition Switch Position w/ Initialize Status
88	PCM	30	XX						Check Engine (MIL) Telltale Indicator Mode Command
0B	PCM	10	XX						Engine Vaccuum Status
1B	PCM	25	XX	XX	XX	XX	XX		Engine RPM w/ High Res. Rate-of-Change w/ Throttle Position Status
3B	PCM	21	XX						Transmission System State Status

TB-6596-A

Figure 6 - Article 02-9-5

Function	Source	DATA							MESSAGE NAME
		1	2	3	4	5	6	7	
3B	PCM	25	XX						Transmission Performance Mode Status
3B	PCM	26	XX						Transmission Operating Mode Status
3B	PCM	65							Request Transmission Performance Mode Command
3B	PCM	83	XX						Transmission Actual Gear Position w/ Shift in Progress Status -Yes (True)
3B	PCM	03	XX						Transmission Actual Gear Position w/ Shift in Progress Status - No (False)
3B	PCM	04	XX						Transmission PRNDL Range Selected Status
3B	PCM	0A	XX						Clutch Pedal Position Status
3B	PCM	3B	XX	XX	XX				Transmission Configuration Status
4B	PCM	10	XX						Engine Oil Fluid Temperature Status
F9	PCM	16	XX	XX					Engine Off Elapsed Time Status
33	REM	22							Brake Lamp Pedal Switch Status - Inactive
33	REM	A2							Brake Lamp Pedal Switch Status - Active
75	REM	41							Request Delayed Accessory Command
75	REM	44							Request Battery Saver Command
83	REM	14	XX	XX	XX	XX	XX		Fuel Level - Sensor A/D Output Status
83	REM	3E	1B	XX	XX				Fuel I/O Status
87	REM	C4							Request Ignition Switch Position w/ Initialize Status
87	REM	C5							Request Key-in-Ignition Status
C4	REM	81	00						All Door Locked Command - Lock(ed)
C4	REM	01	00						All Door Locked Command - Unlock(ed)
C5	REM	A0	26						Passenger's Front Door Lock Switch Status - Active
C5	REM	A1	26						Passenger's Front Door Unlock Switch Status - Active
C7	REM	21	34						Decklid Door Ajar Switch Status - Inactive
C7	REM	21	2A						Driver's Rear Door Ajar Switch Status - Inactive
C7	REM	21	2E						Passenger's Rear Door Ajar Switch Status - Inactive
C7	REM	A1	34						Decklid Door Ajar Switch - Active
C7	REM	A1	2A						Driver's Rear Door Ajar Switch Status - Active
C7	REM	A1	2E						Passenger's Rear Door Ajar Switch Status - Active
C7	REM	E1	26						Request Passenger's Front Door Ajar Switch Status
CB	REM	A1	2E						Passenger's Rear Window Open Switch Status - Active
CB	REM	E3	28						Request All Rear Window Lockout Switch Status
D9	REM	82	39						Left Rear Tail Lamp OK Status - Yes (True)
D9	REM	82	3F						Right Rear Tail Lamp OK Status - Yes (True)
D9	REM	83	39						Left Rear Brake Lamp OK Status - Yes (True)
D9	REM	83	3F						Right Rear Brake Lamp OK Status - Yes (True)
D9	REM	85	39						Left Rear Turn Lamp OK Status - Yes (True)
D9	REM	85	3F						Right Rear Turn Lamp OK Status - Yes (True)
D9	REM	02	39						Left Rear Tail Lamp OK Status - No (False)
D9	REM	02	3F						Right Rear Tail Lamp OK Status - No (False)
D9	REM	03	39						Left Rear Brake Lamp OK Status - No (False)
D9	REM	03	3F						Right Rear Brake Lamp OK Status - No (False)
D9	REM	05	39						Left Rear Turn Lamp OK Status - No (False)
D9	REM	05	3F						Right Rear Turn Lamp OK Status - No (False)
DB	REM	43	38						Request All Rear Brake Lamp Command
DB	REM	44	00						Request All Park Lamp Command
DB	REM	44	38						Request All Rear Park Lamp Command
DB	REM	49	38						Request All Rear Fog Lamp Command
DB	REM	89	38						All Rear Fog Lamp Status - On
DB	REM	09	38						All Rear Fog Lamp Status - Off
E7	REM	81	34						Rear Windshield Electric Defrost Status - On
E7	REM	01	34						Rear Windshield Electric Defrost Status - Off
E7	REM	E1	34						Request Rear Windshield Electric Defrost Status
FE	REM	82							Network Bus Wake-up Command - Yes (True)
8	TC	0C	XX	XX					Desired Axle Torque Command
29	TC	03	XX	XX	XX	XX	XX	XX	Vehicle Speed - Driven and Undriven Wheels - High Res.
33	TC	3B	XX						Brake System Configuration Status
53	TC	FB							Request Engine Configuration Status
88	TC	11							Traction Control Active Telltale Command - Off
88	TC	19							Brake System Failed Telltale Command - Off
88	TC	88							Anti-Lock Brake System Failed Telltale Command - On
88	TC	91							Traction Control Active Telltale Command - On
88	TC	99							Brake System Failed Telltale Command - On
88	TC	08							Anti-Lock Brake System Failed Telltale Command - Off
2B	TC	06	XX	XX					Traction Control System State Status

TB-6597-A

Figure 7 - Article 02-9-5

Function	Source	DATA 1	2	3	4	5	6	7	MESSAGE NAME
3B	TC	C3							Request Transmission Actual Gear Position w/ Shift in Progress Status
3B	TC	FB							Request Transmission Configuration Status
7B	TC	02	XX						Odometer Rolling Count Status
13	TCU	20	XX	XX	XX				Throttle Control Unit Fault Status
33	VACM	E0							Request Parking Brake Switch Status
39	VACM	07							Voice Control Mode Status - Off
39	VACM	87							Voice Control Mode Status - On
96	VACM	91							Chime Configuration 1 Command - Enable(d)
96	VACM	92							Chime Configuration 2 Command - Enable(d)
AB	VACM	C4							Request Emergency Messaging Status
B2	VACM	01	00	XX	XX	XX			All HVAC Operating Mode Command
B2	VACM	0A	00	XX					All HVAC Set Temperature Command
EA	VACM	10	XX	XX	XX	XX	XX	XX	Download Block to Display Command
EA	VACM	11							Terminate Display Definition Command
EA	VACM	30	XX						Display Access Display String Command - Clear Display
EA	VACM	31							Display Access Terminate Command
87	VEMS	C4							Request Ignition Switch Position w/ Initialize Status
AB	VEMS	84							Emergency Messaging Status - Active
AB	VEMS	04							Emergency Messaging Status - Inactive
EA	VEMS	10	XX	XX	XX	XX	XX	XX	Download Block to Display Command
EA	VEMS	11							Terminate Display Definition Command
EA	VEMS	30	XX						Display Access Display String Command - Clear Display
EA	VEMS	31							Display Access Terminate Command

TB-6598-A

Figure 8 - Article 02-9-5